

Psychother Psychosom 2016;85:53–55
DOI: 10.1159/000431257

Group Drumming Modulates Cytokine Response in Mental Health Services Users: A Preliminary Study

Daisy Fancourt^{a, b}, Rosie Perkins^a, Sara Ascenso^a, Louise Atkins^a,
Stephen Kilfeather^c, Livia Carvalho^b, Andrew Steptoe^b,
Aaron Williamson^a

^aCentre for Performance Science, Royal College of Music, and

^bPsychobiology Group, University College, London, and

^cAeirtec Laboratories, Newcastle, UK

Research over the past two decades has explored the role of immune processes in a range of mental health conditions and has shown that pro-inflammatory cytokines, which promote systemic inflammation, can act as neuromodulators, orchestrating immunological and behavioural changes associated with depression [1]. Patients with depression exhibit decreases in cytokine levels proportionate to their degree of recovery when treated either with pharmacotherapy, psychotherapy or a combination of both [2].

In recent years, a number of programmes have developed across the UK and internationally providing psychosocial interventions such as yoga and mindfulness to improve symptoms of mental health conditions and reduce inflammatory response [3, 4]. One promising psychosocial intervention is music, and a recent systematic review identified over 60 studies showing the immune-modulating effects of a range of music interventions [5]. However, to date, music interventions have not been researched as inflammatory-reducing therapeutic agents in mental health.

Consequently, the current pilot study was designed to test the feasibility of using music interventions in mental health. The study explored the effects of group drumming on a broad array of inflammatory measures over a 6-week intervention. Drumming was selected as the music intervention because of the inclusiveness of drumming circles and lack of fine motor skill requirements, and because strong steady rhythms suggest that it may be particularly suitable in mental health settings. Furthermore, two previous short studies (single drumming sessions) involving healthy participants have found changes in biological response [6, 7].

We hypothesised a decrease in depressive symptoms, pro-inflammatory activity and cortisol along with an increase in wellbeing across the entire 6-week intervention, and an increase in positive affect, decrease in cortisol and an increase in immune-enhancing activity evident immediately after a single session.

To test these hypotheses, in the week preceding and following the 6-week intervention, participants completed psychological

scales including the Hospital and Anxiety Depression Scale (HADS), the Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS), the Connor-Davidson Social Resilience Scale (CD-RISC) and the Secker Scale for social inclusion. Immediately before and after the first and final sessions of the 6-week intervention, participants completed visual analogue mood scales, blood pressure measurements and provided a saliva sample, collected via passive drool method and stored immediately at -20°C . Recent research has documented the promise of salivary measurement in psychobiological research as a non-invasive, pain-free tool [8–10]. Saliva samples were analysed using Luminex multiplex assays.

Thirty-one adults with affective disorders accessing mental health services took part in the study (8 men and 23 women; mean age $\pm$ SEM: 52.8 ± 2.45 years), recruited from hospitals, mental health professionals and support organisations and charities operating in North West London. As this was a preliminary study, participants were recruited with a range of mild and moderate mental health conditions. Exclusion criteria were severe anxiety or depression that prevented informed consent, a confounding comorbidity, or use of steroid or immunosuppressive medication. Participants took part in 70-min group drumming sessions in groups of 15–20 over a period of 6 weeks. Sessions consisted of call-and-response exercises and learning drumming patterns that built up into larger pieces.

We used repeated-measures ANOVAs to test changes in subjective reports and immune measures across individual sessions and the whole intervention. Corrections for multiple comparisons were made using Simes' test, and all *p* values presented are corrected. We tested whether age and sex affected the observed differences in outcomes, but since they had no effect we did not control for these variables in the final models.

Results are shown in table 1. Across the entire intervention, significant improvements were found for depression, wellbeing and social resilience. From the beginning to the end of a single session, ratings of stress and tiredness levels significantly decreased and happiness, relaxation and energy levels increased (all *p* < 0.002, data not shown).

Across the entire intervention, the concentrations of four cytokines were significantly lowered, demonstrating a reduction of pro-inflammatory response. There was no evidence of a decrease in cortisol. From the beginning to the end of session 1, the concentration of four cytokines significantly increased, and from the beginning to the end of session 6, seven cytokines significantly increased and cortisol decreased. To test this difference in response noted between week 1 and week 6, we compared change scores calculated by subtracting the scores before a drumming session from scores after the same drumming session. This was only carried out for a subset of participants who had provided viable saliva samples above the level of detection at both time points in sessions

Table 1. Psychological and biological results

	Session 1		F (n)	p ^{b, d}	Session 6		F (n)	p ^b
	before	after			before	after		
<i>Psychological scales (across the 6 weeks, mean ± SD)</i>								
HADSA	9.3±4.2					8.3±3.9 ^d	2.80 (30) ^c	0.105
HADSD	6.3±3.4					4.5±3.1	9.76 (30)	0.007
CD-RISC	60.3±16.5					66.4±17.4	4.70 (31)	0.048
Secker	37.5±7.8					38.4±7.4	0.74 (31)	0.330
WEMWBS	44.6±11.1					49.7±9.0	9.61 (31)	0.007
<i>Biomarkers (across the 6 weeks, mean ± SD)</i>								
IL-2, pg/ml	1.86±0.56 ^c				1.63±0.79		1.00 (14)	0.378
IL-4, pg/ml	0.78±0.58				0.54±0.44		2.85 (16)	0.168
IL-6, pg/ml	0.44±0.36				0.16±0.17		9.94 (15)	0.027
IFN-γ, pg/ml	2.00±0.75				1.34±0.65		9.13 (16)	0.027
TNF-α, pg/ml	1.66±0.38				1.47±0.28		7.01 (18)	0.038
MCP-1, pg/ml	2.78±1.52				2.28±1.49		10.06 (18)	0.027
TGF-β, pg/ml	0.45±0.32				0.40±0.32		0.285 (17)	0.601
Cortisol, nmol/l	4.52±1.82				5.17±1.87		1.87 (18)	0.243
IL-10 ^a	10/15 (67%)				6/15 (40%) ^b			0.168
<i>Biomarkers (across individual sessions, mean ± SD)</i>								
IL-2, pg/ml	1.76±0.64 ^c	1.93±0.79	2.11±26	0.318	1.67±0.81 ^c	2.30±0.78	9.53 (14)	0.016
IL-4, pg/ml	0.62±0.43	0.97±0.68	11.22±24	0.047	0.43±0.19	1.00±0.67	7.57 (12)	0.024
IL-6, pg/ml	0.36±0.31	0.41±0.34	0.63±23	0.639	0.15±0.18	0.44±0.31	12.34 (13)	0.011
IFN-γ, pg/ml	1.63±0.79	1.95±0.89	8.43±24	0.047	1.25±0.68	2.22±0.69	24.12 (13)	0.004
TNF-α, pg/ml	1.57±0.36	1.69±0.43	3.31±25	0.288	1.43±0.25	1.72±0.39	10.71 (16)	0.011
MCP-1, pg/ml	2.41±1.28	2.70±1.41	2.13±25	0.047	2.03±1.31	2.61±1.19	6.08 (14)	0.032
TGF-β, pg/ml	0.40±0.27	0.60±0.47	10.13±27	0.036	0.36±0.29	0.58±0.35	4.13 (15)	0.061
Cortisol, nmol/l	4.74±1.40	4.72±1.72	0.00±28	0.952	5.32±1.72	3.84±2.70	11.00	0.011
IL-10 ^a	17/25 (68%)	16/25 (64%)		0.803	4/13 (31%)	9/13 (69%)		0.024

TNF = Tumour necrosis factor; MCP = monocyte chemotactic protein; TGF-β = transforming growth factor-β; SD = standard deviation.

^a More than 50% of the values of IL-10 were not detectable, so the variable was dichotomised as detectable or not detectable and analysed using McNemar's test.

^b Percent detectable of total.

^c Lower number due to missing data.

^d Data collected the week proceeding the study.

1 and 6. We found that four biomarkers had significantly stronger alterations across session 6 compared with session 1: interleukin (IL)-2 ($p = 0.013$), IL-6 ($p = 0.021$), interferon-γ (IFN-γ, $p < 0.001$) and cortisol ($p = 0.001$). It is possible that responses during the first session were confounded by the anxiety of being in an unusual location with new people doing an unfamiliar activity. Or alternatively, participants may have become more responsive to the intervention as the weeks progressed.

Blood pressure was not reduced although there was a decrease in heart rate of 5.85 bpm ($p = 0.003$). Measures were taken immediately after sessions, and it is possible that greater effects would have emerged over a longer period.

This study demonstrated for the first time that group drumming for mental health service users can lead to reductions in cortisol and immune enhancement over individual sessions as well as reduce inflammatory activity over a 6-week span. Changes in bio-

markers are supported by changes in psychological profiles of participants, demonstrating the potential of group drumming as an intervention for mental health. This supports the undertaking of further controlled studies to test fully the therapeutic potential of group drumming interventions and other music-based psychosocial interventions for mental health patients.

Acknowledgments

This research was carried out as part of 'Creative Practice as Mutual Recovery', a Connected Communities project funded by the UK's Arts and Humanities Research Council (grant ref. AH/K003364/1). The study protocol was approved by the UK NHS National Research Ethics Service under approval ref. 13/LO/1811 and registered under clinical trial number NCT01906892.

References

- 1 Raison CL, Miller AH: Is depression an inflammatory disorder? *Curr Psychiatry Rep* 2011;13:467–475.
- 2 Dahl J, Ormstad H, Aass HCD, Malt UF, Bendz LT, Sandvik L, Brundin L, Andreassen OA: The plasma levels of various cytokines are increased during ongoing depression and are reduced to normal levels after recovery. *Psychoneuroendocrinology* 2014;45:77–86.
- 3 Slavich GM, Irwin MR: From stress to inflammation and major depressive disorder: a social signal transduction theory of depression. *Psychol Bull* 2014;140:774–815.
- 4 Forsman AK, Nordmyr J, Wahlbeck K: Psychosocial interventions for the promotion of mental health and the prevention of depression among older adults. *Health Promot Int* 2011;26:85–107.
- 5 Fancourt D, Ockelford A, Belai A: The psychoneuroimmunological effects of music: a systematic review and a new model. *Brain Behav Immun* 2014;36:15–26.
- 6 Bittman BB, Berk LS, Felten DL, Westengard J, Simonton OC, Pappas J, Ninehouser M: Composite effects of group drumming music therapy on modulation of neuroendocrine-immune parameters in normal subjects. *Altern Ther Health Med* 2001;7:38–47.
- 7 Koyama M, Wachi M, Utsuyama M, Bittman B, Hirokawa K, Kitagawa M: Recreational music-making modulates immunological responses and mood states in older adults. *J Med Dent Sci* 2009;56:79–90.
- 8 Slavish DC, Graham-Engeland JE, Smyth JM, Engeland CG: Salivary markers of inflammation in response to acute stress. *Brain Behav Immun* 2015;44:253–269.
- 9 Williamson S, Munro C, Pickler R, Grap MJ, Elswick RK: Comparison of biomarkers in blood and saliva in healthy adults. *Nurs Res Pract* 2012; 2012:246178.
- 10 Byrne ML, O'Brien-Simpson NM, Reynolds EC, Walsh KA, Loughton K, Waloszek JM, Woods MJ, Trinder J, Allen NB: Acute phase protein and cytokine levels in serum and saliva: a comparison of detectable levels and correlations in a depressed and healthy adolescent sample. *Brain Behav Immun* 2013;34:164–175.